

Supplementary Materials for

A mid-infrared biaxial hyperbolic van der Waals crystal

Zebo Zheng, Ningsheng Xu, Stefano L. Oscurato, Michele Tamagnone, Fengsheng Sun, Yinzhu Jiang, Yanlin Ke, Jianing Chen, Wuchao Huang, William L. Wilson, Antonio Ambrosio*, Shaozhi Deng*, Huanjun Chen*

*Corresponding author. Email: chenhj8@mail.sysu.edu.cn (H.C.); stdsdz@mail.sysu.edu.cn (S.D.); ambrosio@seas.harvard.edu (A.A.)

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(available at advances.sciencemag.org/cgi/content/full/5/5/eaav8690/DC1)

Movie S1 (.mp4 format). Hyperspectral PiFM movie of the edge perpendicular to the [100] direction, which is adjacent to the corner 1 shown in fig. S9.

Movie S2 (.mp4 format). Hyperspectral PiFM movie of the edge perpendicular to the [001] direction, which is in proximity of the corner 1 shown in fig. S9.

Supplementary Notes

Note S1. Theoretical model for calculating the isofrequency surface of the PhPs

Anisotropic permittivities of the α -MoO₃: In the mid-infrared region, the optical response of the α -MoO₃ is dominated by the phonon absorption rather than electronic transition, the dielectric function can be described by the following Lorentz equation (13, 15)

$$\varepsilon_j = \varepsilon_\infty^j \left(1 + \frac{\omega_{LO}^{j\ 2} - \omega_{TO}^{j\ 2}}{\omega_{TO}^{j\ 2} - \omega^2 - i\omega\Gamma^j} \right), \quad j = x, y, z \quad (S1)$$

where ε_j denotes the principal components of the permittivity tensor. The ε_∞^j is the high frequency dielectric constant, the ω_{LO}^j and ω_{TO}^j refer to the longitude (LO) and transverse (TO) optical phonon frequencies, respectively. The parameter Γ^j is the broadening factor of the Lorentzian lineshape. The x, y, and z denote the three principal axes of the crystal, which correspond to the crystalline directions [100], [001], and [010] of the α -MoO₃, respectively. To calculate the relative permittivities shown in Fig. 1A, the phonon frequencies are directly adopted from the literature values (21, 23). The detailed parameters used in our calculation are given in Table S1.

Fresnel equation of biaxial crystal: we start from the Maxwell equations

$$\nabla \cdot \vec{D} = 0 \quad (S2)$$

$$\nabla \cdot \vec{H} = 0 \quad (S3)$$

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad (S4)$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} \quad (S5)$$

where $\vec{D} = \vec{\varepsilon} \cdot \vec{E}$. Considering a monochromatic plane wave propagating in the crystal with a wavevector $\vec{k}$, the corresponding $\vec{E}$, $\vec{D}$, and $\vec{H}$ can be stated as

$$\vec{E} = \vec{E}_0 \exp[i(\vec{k} \cdot \vec{r}) - \omega t] \quad (S6)$$

$$\vec{D} = \vec{D}_0 \exp[i(\vec{k} \cdot \vec{r}) - \omega t] \quad (S7)$$

$$\vec{H} = \vec{H}_0 \exp[i(\vec{k} \cdot \vec{r}) - \omega t] \quad (S8)$$

By substituting the plane wave solution into the Maxwell equations we can obtain the following relations: $\vec{k} \times \vec{E} = \omega \mu_0 \vec{H}_0$, $\vec{k} \times \vec{H} = -\omega \vec{D}$. Note that $\vec{k} = |\vec{k}| \vec{k}_0 = (\omega n/c) \vec{k}_0$ (n denotes the refractive

index of the medium, and $\vec{k}_0$ is the unit vector of the wavevector), and the velocity of light $c = 1/\sqrt{\varepsilon_0\mu_0}$, therefore the electric displacement vector can be determined as

$$\vec{D} = -\frac{1}{\mu_0\omega^2} \vec{k} \times (\vec{k} \times \vec{E}) \quad (\text{S9})$$

$$\vec{D} = -\varepsilon_0 n^2 \vec{k}_0 \times (\vec{k}_0 \times \vec{E}) = \varepsilon_0 n^2 [\vec{E} - \vec{k}_0 (\vec{k}_0 \cdot \vec{E})] \quad (\text{S10})$$

For a biaxial crystal, the three permittivity components are different from each other, $\varepsilon_x \neq \varepsilon_y \neq \varepsilon_z$, which leads to

$$\vec{D}_j = \frac{\varepsilon_0 n^2 \vec{k}_{0j} (\vec{k}_0 \cdot \vec{E})}{\frac{n^2}{\varepsilon_j} - 1} = \frac{\varepsilon_0 \vec{k}_{0j} (\vec{k}_0 \cdot \vec{E})}{\frac{1}{\varepsilon_j} - \frac{1}{n^2}} \quad (\text{S11})$$

The Maxwell equations give $\vec{k} \cdot \vec{D} = 0$, thereby one can obtain

$$\vec{D}_x \vec{k}_{0x} + \vec{D}_y \vec{k}_{0y} + \vec{D}_z \vec{k}_{0z} = 0 \quad (\text{S12})$$

By substituting the Equation S11 into S12, one can get

$$\frac{\varepsilon_0 \vec{k}_{0x}^2 (\vec{k}_0 \cdot \vec{E})}{\frac{1}{\varepsilon_x} - \frac{1}{n^2}} + \frac{\varepsilon_0 \vec{k}_{0y}^2 (\vec{k}_0 \cdot \vec{E})}{\frac{1}{\varepsilon_y} - \frac{1}{n^2}} + \frac{\varepsilon_0 \vec{k}_{0z}^2 (\vec{k}_0 \cdot \vec{E})}{\frac{1}{\varepsilon_z} - \frac{1}{n^2}} = 0 \quad (\text{S13})$$

Finally, we can obtain the Fresnel equation which describes the isofrequency surface of electromagnetic waves within the biaxial crystal as

$$\sum_{j=x,y,z} \frac{\varepsilon_j k_j^2}{n^2 - \varepsilon_j} = 0 \quad (\text{S14})$$

Note S2. Simulation of the dipole-launched PhP waves on the α -MoO₃ surface

To calculate the electric field distributions (Fig. 1E to G) and wavefronts (Fig. 1H to J) of the dipole-launched PhPs, we performed the finite-difference time-domain (FDTD) simulations using a commercial software, the FDTD Solution (Lumerical, Canada). As shown in fig. S2, an electric dipole (frequency range: 500 cm⁻¹ ~ 1250 cm⁻¹) is fixed above the α -MoO₃ flake supported on the SiO₂ substrate. The separation between the dipole source and α -MoO₃ surface is $h = 20$ nm. The thicknesses of the α -MoO₃ and SiO₂ layers are 160 nm and 300 nm, respectively. The anisotropic permittivities of the α -MoO₃ (Equation S1) are imported into the FDTD solutions as user-defined sample data. The magnitudes of the electric field distributions $|E|$ and real parts of the z-components $\text{Re}(E_z)$ at typical frequencies are extracted at the top surface of the α -MoO₃ flake.

Note S3. Simulation of the nanoantenna launched PhPs on the α -MoO₃

The FDTD solution was employed to calculate the wavefronts of the PhPs waves launched by the silver nanoantenna. As shown in fig. S4, a silver nanoantenna is placed onto the surface of the α -MoO₃ flake supported by the SiO₂ substrate. The longitudinal axis of the antenna is parallel to the [100] crystalline direction of the α -MoO₃. The diameter and length of the nanoantenna is 60 nm and 2.5 μ m, respectively. The thicknesses of the α -MoO₃ and SiO₂ layers are 220 nm and 300 nm, respectively. Such a heterostructure is illuminated vertically by a broad-band plane-wave, with polarization parallel to the longitudinal axis of the nanoantenna. The real part of z components of the electric field, $\text{Re}(E_z)$, are obtained from the top surface of the α -MoO₃.

Note S4. Determination of the crystalline direction of a typical α -MoO₃ flake

The orthorhombic α -MoO₃ is a layered oxide composed of distorted and edge-shared MoO₆ octahedra (space group of $Pbnm$) (fig. S5A). The asymmetric oxygen species along the different crystalline directions give rise to the various Mo–O bonding types, resulting in a number of vibration modes that are either IR active or Raman active. As shown in fig. S5C, the Raman peak locating at 158 cm^{-1} depicts the vibration of the translation rigid chain along the [001] direction, and the peak at 820 cm^{-1} is originated from the O–Mo–O stretching vibration along the [100] direction. The peak at 996 cm^{-1} is due to the stretching vibration of the Mo–O bond along the [010] direction (21–23).

To accurately identify the crystalline directions of a specific α -MoO₃ flake, we carried out the polarization-revolved Raman spectroscopy. In a specific measurement, the Raman spectra were consecutively taken upon varying the angles β between the edge I (fig. S5B) of the flake and the polarization direction of the incidence laser. Figure S5C shows the Raman spectra taken with varied excitation polarizations. It is clear that the intensity of the peak at 158 cm^{-1} , which corresponds to the lattice vibration along the [001] direction, is strongly dependent on the excitation polarization. The intensity maximum appears at $\beta = 0^\circ$ where the polarization of the incidence laser is parallel to the longitudinal axis of the flake. As the β is increased, the Raman intensity decreases, which finally disappears at $\beta = 90^\circ$ (Fig. S5D). As a result, the longitudinal direction of the flake is [001] direction, while the transverse one is [100] direction.

Note S5. Simulation of the s-SNOM image using the phenomenological cavity model

The optical near-field distributions on the vdW α -MoO₃ flake can be calculated using a phenomenological cavity model, where the PhPs are launched by a point source (AFM tip) located on the α -MoO₃ surface. The PhPs will propagate radially until reflected by the boundaries. The reflected waves from the boundaries will return to the source point and interfere with the subsequently tip-launched wave. The total field below the point source can be described as (21, 29)

$$\psi = \tilde{\psi}_{\text{PhPs},0} + \sum_j \tilde{\psi}_{\text{PhPs},j} \quad (\text{S15})$$

where $\tilde{\psi}_{\text{PhPs},0}$ is the tip-launched PhPs. The reflected waves from the boundary j can be described as $\tilde{\psi}_{\text{PhPs},j} = R_j \times \tilde{\psi}_{\text{PhPs},0} \exp\left[-(2q)r_j(\gamma + i)\right]$. Parameter $R_j = R_0 \exp(i\Delta\varphi)$ denotes the complex reflection coefficient, which includes the reflectivity R_0 and phase shift $\Delta\varphi$ of the wave. The parameters γ and r_j describe the polariton damping rate and distance between the flake edge and AFM tip, respectively. The interference patterns are associated with $|\psi|$. On the basis of this

model, one can calculate the local field amplitude beneath the AFM tip at arbitrary position on the α -MoO₃ sheet.

In the s-SNOM measurement, the AFM tip not only excites the PhPs by focusing the incidence mid-infrared light onto the α -MoO₃ flake, but also acts as a near-field detector for recording the total field beneath the tip. In the calculations the amplitude of the tip-launched polariton $\tilde{\psi}_{PhPs,0}$ was set to 1. For calculating the image shown in the Fig. 3B, the damping rate γ was fixed at 0.12 and the values of q can be calculated through the following equation $q(\omega) + iq'(\omega) = \frac{2}{d \tan \theta} \Delta\phi$

(13, 15), where $\Delta\phi = \arctan(\frac{\varepsilon_0}{\varepsilon_{\perp} \tan \theta}) + \arctan(\frac{\varepsilon_{sub}}{\varepsilon_{\perp} \tan \theta})$. Note that the in-plane wavevector q is strongly dependent on the propagation direction. To calculate the q along the [001] direction ([100] direction), the $\varepsilon_{\perp}$ is substituted by ε_y (ε_x). The ε_{sub} denotes the relative permittivity of the SiO₂ substrate. For the complex reflection coefficient, the R_0 and $\Delta\phi$ were set as 1 and 1.5π , respectively.

Note S6. Calculations on the complex reflectivity of the multilayer structure α -MoO₃/SiO₂

The PhPs dispersion relation of the α -MoO₃ flake can be manifested by the complex reflectivity $r_p(\omega, q)$ of a multilayer structure consisted of air/ α -MoO₃/SiO₂ (fig. S8). The total reflectivity of this system is determined by Equation S16 (13)

$$r_p = \frac{r_1 + r_3 e^{i2k_2^b d}}{1 + r_1 r_3 e^{i2k_2^b d}} \quad (S16)$$

$$r_1 = \frac{\varepsilon_{\perp} k_1^b - \varepsilon_1 k_2^b}{\varepsilon_{\perp} k_1^b + \varepsilon_1 k_2^b} \quad (S17)$$

$$r_3 = \frac{\varepsilon_3 k_2^b - \varepsilon_{\perp} k_3^b}{\varepsilon_3 k_2^b + \varepsilon_{\perp} k_3^b} \quad (S18)$$

where the subscripts “1”, “2”, and “3” denote the air, α -MoO₃, and SiO₂, respectively. The parameter $d = 160$ nm is the thickness of the α -MoO₃ flake. The parameters r_1 and r_2 delegate the reflectivities at the air/ α -MoO₃ and α -MoO₃/SiO₂ interfaces, respectively. The ε_j ($j = 1, 3$, and $\perp$) refers to the relative permittivities of the air, SiO₂, and the component perpendicular to the z-axis ([010] direction) of the α -MoO₃ flake. To calculate the dispersion relation of the PhPs propagating along the [001] direction ([100] direction), the $\varepsilon_{\perp}$ is substituted by ε_y (ε_x), which are obtained from the Equation S1. The parameter k_j^b is the momentum component of the

propagating wave along the b axis. For $j = 1$ and 3, the momenta are given by $k_j^b = \sqrt{\varepsilon_j \left(\frac{\omega}{c}\right)^2 - q^2}$

, while for the electromagnetic wave in the anisotropic α -MoO₃ $k_2^b = \sqrt{\varepsilon_{\perp} \left(\frac{\omega}{c}\right)^2 - \frac{\varepsilon_{\perp}}{\varepsilon_{//}} q^2}$.

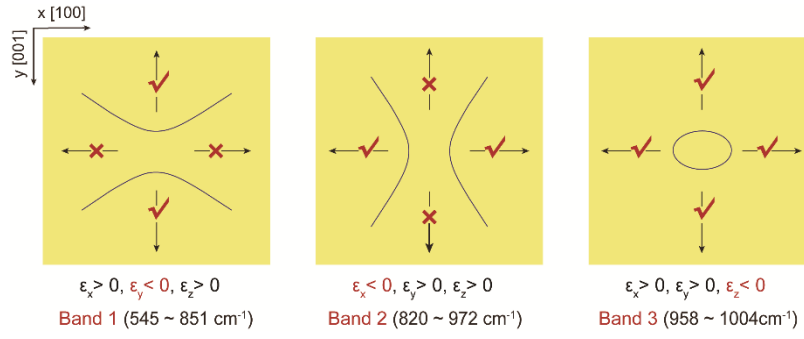


Fig. S1. Schematic showing the isofrequency curves of the PhPs in the biaxial $\alpha\text{-MoO}_3$ flake.

The opening directions of the hyperbolic isofrequency curves can be tuned by varying the illumination frequencies within the Band 1 (left) and Band 2 (middle), indicating the PhPs can propagate along the direction where the permittivity is negative. The right panel shows conventional in-plane anisotropic PhPs with elliptical wavefront, indicating the PhPs can propagate in all directions, but with different wavevectors.

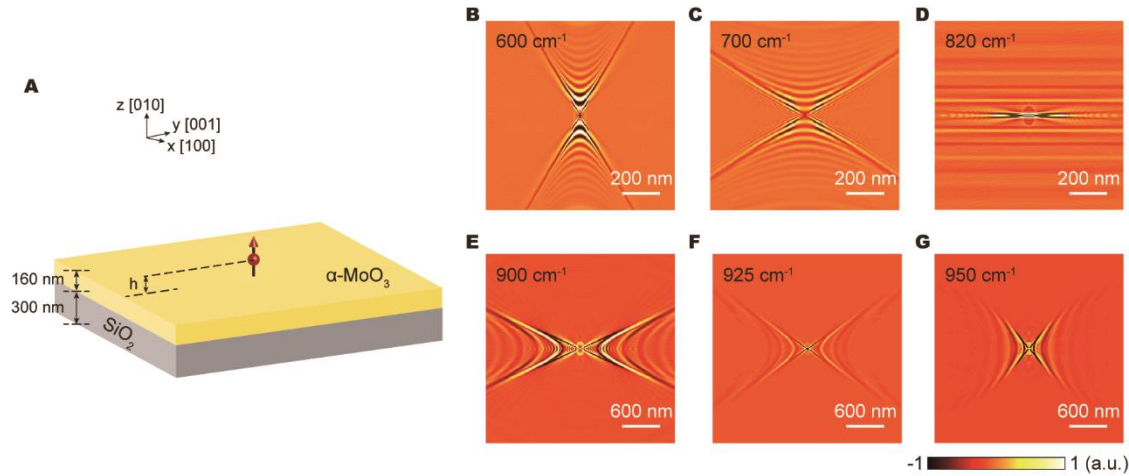


Fig. S2. Simulations of the dipole-launched hyperbolic PhPs on the $\alpha\text{-MoO}_3$ surface. (A) Calculation scheme. The thicknesses of the $\alpha\text{-MoO}_3$ flake and SiO_2 substrate are 160 nm and 300 nm, respectively. The $h = 20$ nm denotes the separation between the electric dipole and $\alpha\text{-MoO}_3$ surface. (B–G) In-plane hyperbolic PhPs with different wavefronts and propagation directions at various excitation frequencies.

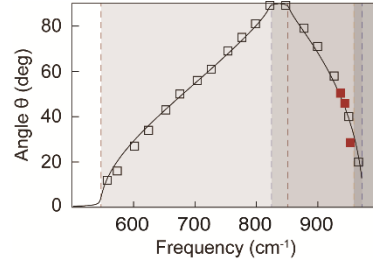


Fig. S3. Propagation directions of the in-plane hyperbolic PhPs. The solid line plots the theoretical calculations using the equation $\theta(\omega) = \frac{\pi}{2} - \arctan\left(\frac{\sqrt{\epsilon_y(\omega)}}{i\sqrt{\epsilon_x(\omega)}}\right)$, the hollow squares are extracted data from the FDTD simulations, and the red solid squares are data obtained from the near-field optical images shown in Fig. 2.

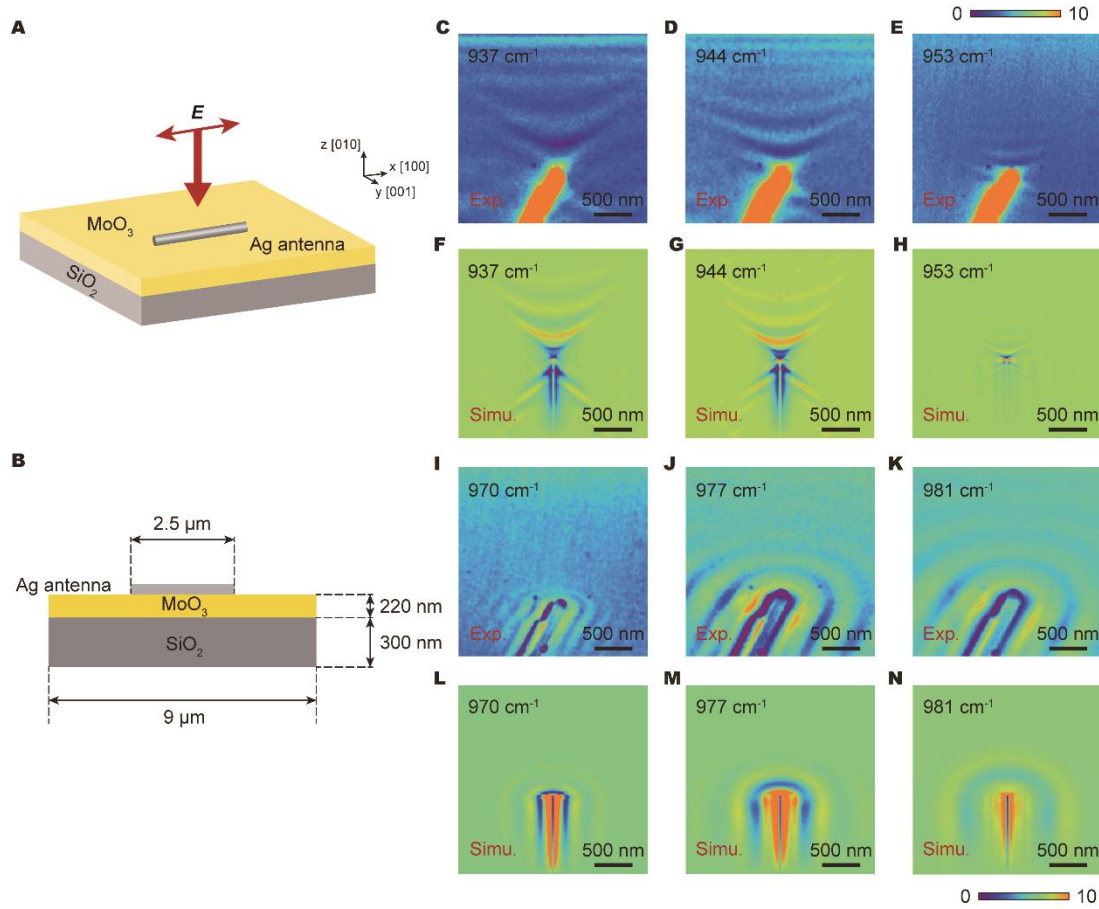


Fig. S4. Silver nanoantenna-launched PhPs at various incidence frequencies. (A) Schematic showing the simulation geometry. (B) Side-view of the nanoantenna/ α -MoO₃/SiO₂ heterostructure. (C–E) Near-field optical images of the in-plane hyperbolic PhPs launched by the silver nanoantenna. The illumination frequencies locate in the Band 2. (F–H) Simulated electric field distributions of the nanoantenna-launched PhPs, which correspond to (C)–(E). (J–K) Near-field optical images of the PhPs excited with frequencies in the Band 3. (L–N) Simulated electric field distributions of the nanoantenna-launched PhPs, which correspond to (J) – (K).

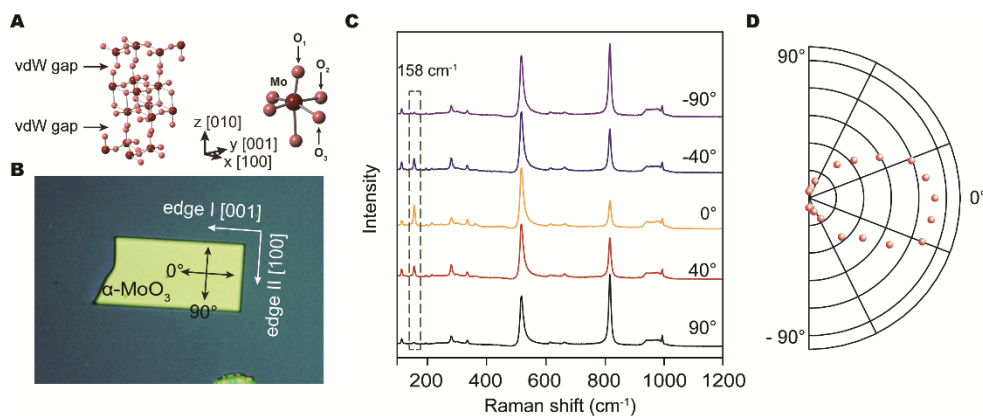


Fig. S5. Determination of the crystalline directions of the α -MoO₃ by Raman spectroscopy. (A) Crystalline structure of the α -MoO₃ and the corresponding octahedron unit cell. (B) Optical image of a typical α -MoO₃ flake with two orthogonal edges, where one of them is along the [001] direction (edge I) and the other is along the [100] direction (edge II). The black double-headed arrows indicate the polarization directions of the incidence laser, which is defined by the angle β with respect to the [001] direction. (C) The Raman spectra of the α -MoO₃ obtained at different β . (D) Polar plots of the Raman intensities at 158 cm⁻¹.

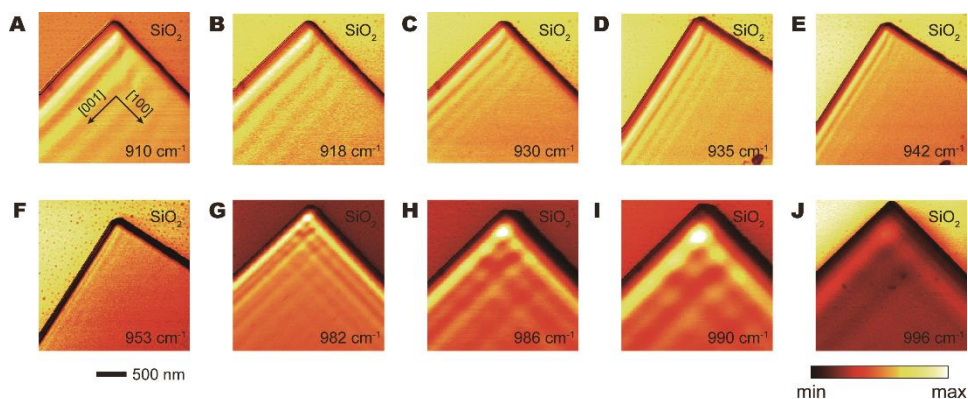


Fig. S6. Near-field optical images showing in-plane anisotropic PhPs characteristics of the α -MoO₃. (A–F) Near-field optical images obtained with different illumination frequencies in the Band 2, where the in-plane hyperbolic responses take place. (G–J) Near-field optical images obtained in the Band 3 without the hyperbolic responses.

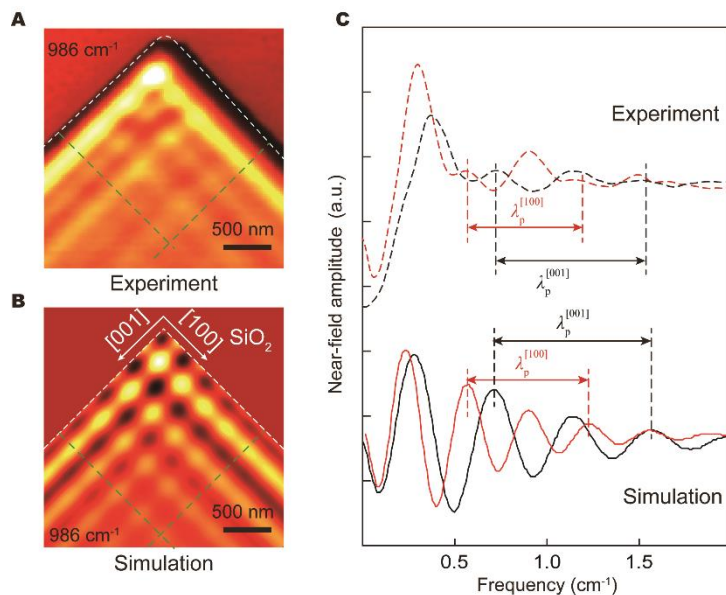


Fig. S7. Comparison of the experiment and simulation near-field images of the PhPs distributions illuminated by 986 cm^{-1} (Band 3). (A) Near-field optical image obtained using the s-SNOM. (B) Simulated image using the phenomenological cavity model described in the note S7. (C) Comparison of the extracted profiles along the green dash lines marked in (A) and (B).



Fig. S8. Scheme of the multi-layered structure consisted of air/ $\alpha\text{-MoO}_3$ / SiO_2 . The characters “1”, “2”, and “3” denote the air, $\alpha\text{-MoO}_3$, and SiO_2 , respectively.

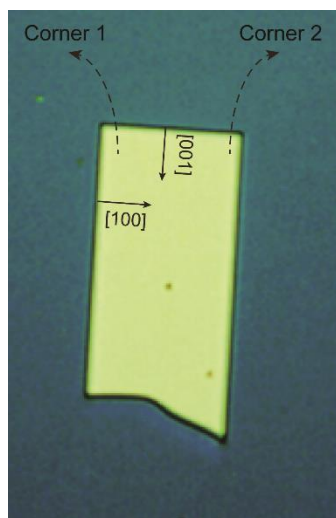


Fig. S9. Optical image of the $\alpha\text{-MoO}_3$ flake used for conducting the hyperspectral PiFM movies.

Movie S1. Hyperspectral PiFM movie of the edge perpendicular to the [100] direction, which is adjacent to the corner 1 shown in fig. S9. The scan size is $3\text{ }\mu\text{m} \times 3\text{ }\mu\text{m}$. The spectral range is $776\text{ cm}^{-1} \sim 1000\text{ cm}^{-1}$.

Movie S2. Hyperspectral PiFM movie of the edge perpendicular to the [001] direction, which is in proximity of the corner 1 shown in fig. S9. The scan size is $2\text{ }\mu\text{m} \times 2\text{ }\mu\text{m}$. The spectral range is $776\text{ cm}^{-1} \sim 1000\text{ cm}^{-1}$.

Table S1. Parameters used in calculating the relative permittivities (Eq. S1).

Direction	x [100]	x [100]	y [001]	y [001]	z [010]	y [010]
	Ref. (20, 21)	this work	Ref. (20, 21)	this work	Ref. (20, 21)	this work
ε_{∞}	\	4.0	\	5.2	\	2.4
$\omega_{LO}/\text{cm}^{-1}$	974	972	851	851	1010	1004
$\omega_{TO}/\text{cm}^{-1}$	818	820	545	545	962	958
Γ/cm^{-1}	\	4	\	4	\	2